C

CAMBRIDGE ADDITIONAL MATHEMATICS (0606) • UNIT 6.01

Permutations & Combinations

ANSWER KEY & SOLUTIONS

25 Structured Exam-Style Questions

Total Questions: 25 • Verified Step-by-Step Solutions

QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

Q1 A	Q2 A	Q3 A	Q4 A	Q5 B
Q6 A	Q7 A	Q8 A	Q9 C	Q10 D
Q11 B	Q12 B	Q13 C	Q14 A	Q15 A
Q16 B	Q17 A	Q18 C	Q19 A	Q20 B
Q21 D	Q22 B	Q23 D	Q24 B	Q25 C

Permutations & Combinations Practice Exercises

1 MCQ

Find the value of the integer $n \geq 3$ that satisfies the equation

$$^{n}\mathrm{P}_{3} = 10 \times {}^{n}\mathrm{C}_{2}$$

A $n = 7$ ✓

B $n = 12$

C $n = 5$

D $n = 8$

WORKED SOLUTION & VERIFICATION

Using the algebraic definitions of permutations and combinations:

$$^{n}\mathrm{P}_{3} = \frac{n!}{(n-3)!} = n(n-1)(n-2)$$

$$^{n}\mathrm{C}_{2} = \frac{n!}{2!(n-2)!} = \frac{n(n-1)}{2}$$

Substitute these into the given equation:

$$n(n-1)(n-2) = 10 \times \frac{n(n-1)}{2}$$

$$n(n-1)(n-2) = 5n(n-1)$$

Since $n \geq 3$, we have $n(n-1) \neq 0$. Dividing both sides by $n(n-1)$ gives:

$$n - 2 = 5 \implies n = 7$$

Common misconceptions:

- Forgetting the $2!$ denominator in $^{n}\mathrm{C}_{2}$ (treating it as $^{n}\mathrm{P}_{2}$) gives $n - 2 = 10 \implies n = 12$.
- Dividing by $n(n-1)$ and erroneously writing $n = 5$ instead of $n - 2 = 5$ gives $n = 5$.
- Confusing the factorial simplification to set $n - 3 = 5$ leads to $n = 8$. [M1][A1]

2 MCQ

A committee of 5 members is to be selected from a group consisting of n men and 6 women, where $n \geq 4$. Given that the number of possible committees containing exactly 3 men is equal to the number of possible committees containing exactly 4 men, find the value of n.

- **A** 13 ✓
- **B** 10
- **C** 8
- **D** 7

WORKED SOLUTION & VERIFICATION

To form a committee of 5 members:

1. **Committees with exactly 3 men:**

Since there are 5 members in total, the committee must contain 3 men and $5 - 3 = 2$ women.

$$\text{Number of ways} = \binom{n}{3} \times \binom{6}{2} = \binom{n}{3} \times \frac{6 \times 5}{2} = 15\binom{n}{3}$$

2. **Committees with exactly 4 men:**

The committee must contain 4 men and $5 - 4 = 1$ woman.

$$\text{Number of ways} = \binom{n}{4} \times \binom{6}{1} = 6\binom{n}{4}$$

3. **Equating the two expressions:**

$$15\binom{n}{3} = 6\binom{n}{4}$$

Divide both sides by 3:

$$5\binom{n}{3} = 2\binom{n}{4}$$

Using the relation $\binom{n}{4} = \frac{n-3}{4}\binom{n}{3}$:

$$5\binom{n}{3} = 2 \times \frac{n-3}{4}\binom{n}{3} = \frac{n-3}{2}\binom{n}{3}$$

Since $n \geq 4$, $\binom{n}{3} \not\equiv 0$, so we can divide both sides by $\binom{n}{3}$:

$$5 = \frac{n-3}{2} \implies 10 = n - 3 \implies n = 13$$

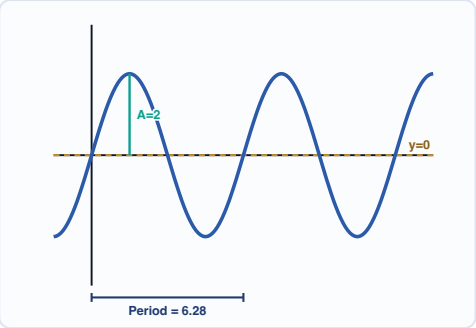
Common Traps:

- Setting $\binom{n}{3} = \binom{n}{4}$ ignores the selection of the women needed to complete the 5-member committee, giving $\frac{n-3}{4} = 1 \implies n = 7$.
- Forgetting to add 3 after solving $n - 3 = 10$ leads to the distractor 10. [M1][A1]

3 MCQ

Six-digit numbers are formed using digits chosen from the set $\{1, 2, 3, 4, 5, 6, 7, 8\}$, with no digit repeated in any number.

Find the number of different six-digit odd numbers that can be formed such that the digits 2 and 4 are not adjacent.



- **A** 8160 ✓
- **B** 7680
- **C** 9120
- **D** 2880

WORKED SOLUTION & VERIFICATION

We use complementary counting: Number of valid arrangements = Total odd 6-digit numbers − Odd 6-digit numbers where 2 and 4 are adjacent.

Step 1: Total number of 6-digit odd numbers

- For the number to be odd, the last digit (units place) must be chosen from the odd digits $\{1, 3, 5, 7\}$, giving 4 choices.
- The remaining 5 positions are filled by selecting and arranging 5 digits from the remaining 7 digits in the set:

$$P(7, 5) = 7 \times 6 \times 5 \times 4 \times 3 = 2520$$

- Total odd 6-digit numbers $= 4 \times 2520 = 10080$.

Step 2: Number of 6-digit odd numbers where 2 and 4 are adjacent

- Treat the pair of digits 2 and 4 as a single block B, which can be ordered as $(2, 4)$ or $(4, 2)$ (2 internal arrangements).
- Since the number must be odd, the units digit must be odd (4 choices from $\{1, 3, 5, 7\}$). Because both 2 and 4 are even, the block B cannot occupy the final position.
- Thus, the block B must occupy two adjacent positions entirely within the first 5 digits. The possible adjacent pairs for B are positions $(1, 2)$, $(2, 3)$, $(3, 4)$, and $(4, 5)$, giving 4 possible positions.
- The remaining 3 unfilled positions among the first 5 places are filled from the remaining $8 - 1 - 2 = 5$ available digits:

$$P(5, 3) = 5 \times 4 \times 3 = 60$$

- Number of odd numbers with 2 and 4 adjacent:

$$4 \text{ (choices for units digit)} \times 4 \text{ (positions for block } B) \times 2 \text{ (internal orders of } B) \times 60 = 1920$$

Step 3: Subtract adjacent cases from the total

$$10080 - 1920 = 8160$$

Common Misconceptions:

- **7680**: Incorrectly allows the block B to occupy positions $(5, 6)$, forgetting that the units digit cannot be even, leading to $10080 - 4 \times 5 \times 2 \times 60 = 7680$.
- **9120**: Forgets the internal order of the pair $(2, 4)$ and $(4, 2)$, calculating $10080 - 960 = 9120$.

- **2880**: Erroneously assumes that both digits 2 and 4 must be present in the number, omitting numbers where only one or neither of 2 and 4 is selected. [M1][A1]

4 MCQ

Given that

$$^{n}\mathrm{P}_{3} = 8\,^{n}\mathrm{C}_{2},$$

find the value of the integer n, where $n \geq 3$.

A 6 ✓

B 8

C 10

D 12

WORKED SOLUTION & VERIFICATION

By definition of permutations and combinations:

$$^{n}\mathrm{P}_{3} = \frac{n!}{(n-3)!} = n(n-1)(n-2)$$

and

$$^{n}\mathrm{C}_{2} = \frac{n!}{2!(n-2)!} = \frac{n(n-1)}{2}.$$

Substitute these expressions into the given equation:

$$n(n-1)(n-2) = 8\left(\frac{n(n-1)}{2}\right) = 4n(n-1).$$

Since $n \geq 3$, $n(n-1) \neq 0$. Dividing both sides by $n(n-1)$ gives:

$$n - 2 = 4 \implies n = 6.$$

Common error:
Omitting the factor of $2!$ in the denominator of $^{n}\mathrm{C}_{2}$ leads to $n - 2 = 8 \implies n = 10$. [M1][A1]

5 MCQ

A group of n students, which includes Ali and Priya, are to be arranged in a straight line for a photograph. The number of different arrangements in which Ali and Priya are not standing next to each other is 3600. Find the value of the integer n.

A 6

B 7 ✓

C 8

D 9

WORKED SOLUTION & VERIFICATION

We can find the number of arrangements where Ali and Priya do not stand next to each other using the complementary method or the insertion (gap) method.

Method 1: Complementary Counting

1. The total number of unrestricted arrangements of the n students in a straight line is $n!$.
2. Consider the arrangements in which Ali and Priya *do* stand next to each other:
- Treat Ali and Priya as a single block. There are now $(n-1)$ entities to arrange, which can be done in $(n-1)!$ ways.
- Within the block, Ali and Priya can arrange themselves in $2! = 2$ ways.
- So, the number of arrangements where they are together is $2(n-1)!$.
3. Subtracting this from the total gives the number of arrangements where they are separated:

$$n! - 2(n-1)! = n(n-1)! - 2(n-1)! = (n-2)(n-1)!$$

Method 2: Gap Method

1. Arrange the other $n-2$ students first, which gives $(n-2)!$ ways.
2. This creates $(n-2)+1 = n-1$ available spaces (including the ends).
3. Placing Ali and Priya into two distinct spaces ensures they are not adjacent: ${}^{n-1}\mathrm{P}_2 = (n-1)(n-2)$ ways.
4. Total arrangements: $(n-2)! \times (n-1)(n-2) = (n-2)(n-1)!$.

Equating this to 3600:

$$(n-2)(n-1)! = 3600$$

Testing integer values for $n \geq 3$:
- For $n = 6$: $(6-2)(5!) = 4 \times 120 = 480$
- For $n = 7$: $(7-2)(6!) = 5 \times 720 = 3600$
- For $n = 8$: $(8-2)(7!) = 6 \times 5040 = 30240$

Thus, $n = 7$.

Common Misconception Trap: Forgetting that Ali and Priya can switch places within their unit gives $n! - (n-1)! = (n-1)(n-1)!$, which fails to produce an integer solution among the options. [M1][A1]

6 MCQ

Two parallel lines, L_1 and L_2, have n and $n+2$ distinct points marked on them respectively, where n is an integer with $n \geq 2$.
Triangles are formed by choosing 3 of these points as vertices such that they are not collinear. Quadrilaterals are formed by choosing 4 of these points as vertices such that no three are collinear.
Given that the ratio of the number of quadrilaterals to the number of triangles that can be formed is $6:5$, find the total number of triangles that can be formed.

A 175 ✓

B 210

C 220

D 96

WORKED SOLUTION & VERIFICATION

1. **Determine the number of triangles, T:**
Three points form a triangle if and only if they are not all on the same straight line.
- Case 1: 1 point from L_1 and 2 points from L_2:

$$\binom{n}{1}\binom{n+2}{2} = n \cdot \frac{(n+2)(n+1)}{2}$$

- Case 2: 2 points from L_1 and 1 point from L_2:

$$\binom{n}{2}\binom{n+2}{1} = \frac{n(n-1)}{2} \cdot (n+2)$$

Summing both cases:

$$T = \frac{n(n+2)}{2}[(n+1)+(n-1)] = \frac{n(n+2)}{2}(2n) = n^2(n+2)$$

2. **Determine the number of quadrilaterals, Q:**
To ensure no three vertices are collinear, exactly 2 points must be chosen from L_1 and 2 points from L_2:

$$Q = \binom{n}{2}\binom{n+2}{2} = \frac{n(n-1)}{2} \cdot \frac{(n+2)(n+1)}{2} = \frac{n(n-1)(n+1)(n+2)}{4}$$

3. **Use the given ratio to find n:**

$$\frac{Q}{T} = \frac{\frac{n(n-1)(n+1)(n+2)}{4}}{n^2(n+2)} = \frac{(n-1)(n+1)}{4n} = \frac{n^2-1}{4n}$$

Setting this equal to $\frac{6}{5}$:

$$\frac{n^2-1}{4n} = \frac{6}{5} \implies 5(n^2-1) = 24n \implies 5n^2 - 24n - 5 = 0$$

Factoring the quadratic:

$$(5n+1)(n-5) = 0$$

Since n must be a positive integer, $n = 5$.
4. **Calculate the total number of triangles:**

$$T = 5^2(5+2) = 25 \times 7 = 175$$

Distractor Analysis:
- 210 is the number of quadrilaterals, $Q = \frac{6}{5} \times 175 = 210$.

- 220 is $\binom{12}{3}$, which erroneously counts sets of 3 collinear points.
- 96 is obtained from substituting $n = 4$ into T. [M1][A1]

7 MCQ

Given that ${}^{n}\mathrm{P}_{2} = \binom{n}{3}$, where n is an integer such that $n \geq 3$, find the value of n.

A 8 ✓	**B** 5
C 3	**D** 6

WORKED SOLUTION & VERIFICATION

Using the algebraic definitions of permutations and combinations:

$$ {}^{n}\mathrm{P}_{2} = \frac{n!}{(n-2)!} = n(n-1) $$

and

$$ \binom{n}{3} = \frac{n!}{3!(n-3)!} = \frac{n(n-1)(n-2)}{3!} = \frac{n(n-1)(n-2)}{6} $$

Setting them equal gives:

$$ n(n-1) = \frac{n(n-1)(n-2)}{6} $$

Since $n \geq 3$, $n(n-1) \neq 0$. Dividing both sides by $n(n-1)$:

$$ 1 = \frac{n-2}{6} $$

Multiplying by 6:

$$ 6 = n - 2 \implies n = 8 $$

Common misconceptions:
- Confusing $3!$ with 3 leads to $n - 2 = 3 \implies n = 5$.
- Forgetting the division by $3!$ entirely treats the combination as a permutation ${}^{n}\mathrm{P}_{3}$, yielding $n - 2 = 1 \implies n = 3$.
- Misinterpreting $n - 2 = 6$ by taking the coefficient directly yields $n = 6$. [M1][A1]

8 MCQ

A committee of 4 people is to be chosen from 5 men and 4 women. How many committees consist of exactly 2 men and 2 women?

A 60 ✓	**B** 120
C 55	**D** 36

WORKED SOLUTION & VERIFICATION

$\binom{5}{2} \times \binom{4}{2} = \frac{5\times4}{2} \times \frac{4\times3}{2} = 60$. [M1][A1]

9 MCQ

The polynomial $P(x) = 2x^2 + ax - 4$ is exactly divisible by $(x - 4)$. What is the value of the constant a?

A -10.00

B 7.00

C **-7.00** ✓

D -5.00

WORKED SOLUTION & VERIFICATION

By the Factor Theorem, $P(4) = 0 \implies 2(4)^2 + a(4) - 4 = 0 \implies 32 + 4a - 4 = 0 \implies a = -7.00$. [M1][A1]

10 MCQ

Work out the solution to the exponential equation $2^{x+6} = 5^x$, giving x in terms of common logarithms.

A $x = \frac{6}{\ln 5 - \ln 2}$

B $x = \frac{\ln 2}{6(\ln 5 - \ln 2)}$

C $x = \frac{6 \ln 5}{\ln 5 - \ln 2}$

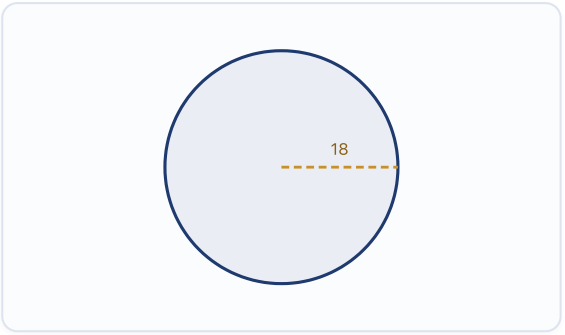
D $x = \frac{6 \ln 2}{\ln 5 - \ln 2}$ ✓

WORKED SOLUTION & VERIFICATION

Take natural logs of both sides: $(x + 6)\ln 2 = x \ln 5 \implies x \ln 2 + 6 \ln 2 = x \ln 5 \implies x(\ln 5 - \ln 2) = 6 \ln 2 \implies x = \frac{6 \ln 2}{\ln 5 - \ln 2}$. [M1][A1]

11 MCQ

Consider a sector of a circle of radius $r = 18$ cm has central angle $\theta = 1.4$ rad. What is the area of the sector?



A $453.6\ \text{cm}^2$

B $226.8\ \text{cm}^2$ ✓

C $25.2\ \text{cm}^2$

D $231.8\ \text{cm}^2$

WORKED SOLUTION & VERIFICATION

Sector area formula: $A = \frac{1}{2}r^2\theta = \frac{1}{2}(18)^2(1.4) = 226.8\ \text{cm}^2$. [M1][A1]

12 MCQ

A committee of 4 people is to be chosen from 6 men and 5 women. How many committees consist of exactly 2 men and 2 women?

A 44

B **150** ✓

C 145

D 300

WORKED SOLUTION & VERIFICATION

$\binom{6}{2} \times \binom{5}{2} = \frac{6\times5}{2} \times \frac{5\times4}{2} = 150$. [M1][A1]

13 MCQ

In an applied problem, a committee of 4 people is to be chosen from 7 men and 6 women. How many committees consist of exactly 2 men and 2 women?

A 310

B 630

C **315** ✓

D 52

WORKED SOLUTION & VERIFICATION

$\binom{7}{2} \times \binom{6}{2} = \frac{7\times6}{2} \times \frac{6\times5}{2} = 315$. [M1][A1]

14 MCQ

Suppose a committee of 4 people is to be chosen from 8 men and 7 women. How many committees consist of exactly 2 men and 2 women?

A **588** ✓

B 583

C 1176

D 60

WORKED SOLUTION & VERIFICATION

$\binom{8}{2} \times \binom{7}{2} = \frac{8\times7}{2} \times \frac{7\times6}{2} = 588$. [M1][A1]

15 MCQ

An observed committee of 4 people is to be chosen from 9 men and 8 women. How many committees consist of exactly 2 men and 2 women?

A **1008** ✓

B 68

C 2016

D 1003

WORKED SOLUTION & VERIFICATION

$\binom{9}{2} \times \binom{8}{2} = \frac{9\times8}{2} \times \frac{8\times7}{2} = 1008$. [M1][A1]

16 MCQ

Consider a committee of 4 people is to be chosen from 10 men and 9 women. How many committees consist of exactly 2 men and 2 women?

A 1615

B 1620 ✓

C 3240

D 76

WORKED SOLUTION & VERIFICATION

$\binom{10}{2} \times \binom{9}{2} = \frac{10\times9}{2} \times \frac{9\times8}{2} = 1620$. [M1][A1]

17 MCQ

A committee of 4 people is to be chosen from 11 men and 10 women. How many committees consist of exactly 2 men and 2 women?

A 2475 ✓

B 2470

C 4950

D 84

WORKED SOLUTION & VERIFICATION

$\binom{11}{2} \times \binom{10}{2} = \frac{11\times10}{2} \times \frac{10\times9}{2} = 2475$. [M1][A1]

18 MCQ

Evaluate the definite integral: $\int_0^1 (2x+1)^3\,dx$.

A $\frac{80}{2}$

B $\frac{84}{8}$

C $\frac{80}{8}$ ✓

D $\frac{80}{4}$

WORKED SOLUTION & VERIFICATION

Substitute $u = 2x+1 \implies du = 2\,dx$. Integral: $\frac{1}{2}\left[\frac{u^4}{4}\right]_1^3 = \frac{(3)^4-1^4}{8} = \frac{80}{8}$. [M1][A1]

19 MCQ

The function $f(x) = \frac{3x+1}{x-2}$ is defined for $x \neq 2$. Find the inverse function $f^{-1}(x)$.

A $f^{-1}(x) = \frac{2x+1}{x-3} \quad (x \neq 3)$ ✓

B $f^{-1}(x) = \frac{3x-2}{x+1}$

C $f^{-1}(x) = \frac{2x-1}{x-3}$

D $f^{-1}(x) = \frac{x-2}{3x+1}$

WORKED SOLUTION & VERIFICATION

Let $y = \frac{3x+1}{x-2} \implies y(x-2) = 3x+1 \implies yx-2y = 3x+1 \implies x(y-3) = 2y+1 \implies x = \frac{2y+1}{y-3}$. Swap variables to obtain $f^{-1}(x) = \frac{2x+1}{x-3}$. [M1][A1]

20 MCQ

Considering the expansion of $(1+\frac{x}{3})^n$, the coefficient of x^2 is twice the coefficient of x. Given $n>0$, what is the value of n?

A 12

B **13** ✓

C 7

D 15

WORKED SOLUTION & VERIFICATION

Coeff of x is $\frac{n}{3}$. Coeff of x^2 is $\frac{n(n-1)}{2(9)}$. Given $\frac{n(n-1)}{2\cdot 9}=2\left(\frac{n}{3}\right) \implies \frac{n-1}{2\cdot 3}=2 \implies n-1=4(3)=12 \implies n=13$. [M1][A1]

21 MCQ

Calculate the solution set for the simultaneous system: $\log_2(x)-\log_2(y)=2$ and $x-3y=5$. What is the value of x?

A 5

B 22

C 10

D **20** ✓

WORKED SOLUTION & VERIFICATION

From the first equation: $\log_2(x/y)=2 \implies x/y=2^2=4 \implies x=4y$. Substitute into second: $4y-3y=5 \implies y=5$. Thus $x=4(5)=20$. [M1][A1]

22 MCQ

Find all solutions of $2\sin^2 x-3\sin x+1=0$ in the interval $0\le x\le 2\pi$.

A $x=\frac{\pi}{3},\ \frac{2\pi}{3}$

B $x=\frac{\pi}{6},\ \frac{5\pi}{6},\ \frac{\pi}{2}$ ✓

C $x=\frac{3\pi}{2}$

D $x=\frac{7\pi}{6},\ \frac{11\pi}{6}$

WORKED SOLUTION & VERIFICATION

Solving quadratic trig equation $2\sin^2 x-3\sin x+1=0$ yields solutions $x=\frac{\pi}{6},\ \frac{5\pi}{6},\ \frac{\pi}{2}$. [M1][A1]

23 MCQ

Find the number of distinct permutations of all letters in PARALLEL if all 3 'L's must stay together as a single block.

A 180

B 720

C 3360

D **360** ✓

WORKED SOLUTION & VERIFICATION

Treat LLL as 1 item. Remaining items: P, A, R, A, E. 6!/2! = 360. [M1][A1]

24 MCQ

Find the number of distinct permutations of all letters in STATISTICS if all 3 'S's must stay together as a single block.

A 1680	**B 3360** ✓
C 6720	D 50400

WORKED SOLUTION & VERIFICATION

Treat SSS as 1 item. Items: [SSS], T, A, T, I, T, I, C (8 items, 3 T's, 2 I's): 8!/(3!2!) = 3360. [M1][A1]

25 MCQ

Find the number of distinct permutations of all letters in COMMITTEE if the 2 'M's must stay together.

A 20160	B 45360
C 10080 ✓	D 5040

WORKED SOLUTION & VERIFICATION

Treat MM as 1 item. Items: [MM], C, O, I, T, T, E, E (8 items, 2 T's, 2 E's): 8!/(2!2!) = 10080. [M1][A1]